LAB 5 — COMPREHENSIVE THREAT INVESTIGATION

Dataset scope: index="soc-investigation", source="*05_Comprehensive_Threat_Investigation_Raw.csv*". The CSV is simulated telemetry, not genuine endpoint logs or a packet capture.

Investigation Searches

01 - Incident Timeline

Purpose: Incident Timeline.

Splunk SPL

```
index="soc-investigation" source="*05_Comprehensive_Threat_Investigation_Raw.csv*"
| sort 0 _time
| table _time ComputerName User EventCode ProcessName ParentProcessName CommandLine DestinationIp
DestinationPort TargetFilename Details
```

Explanation: Builds a chronological view of the Lab 5 events. The source filter limits results to the simulated CSV; sort orders events by time; table displays fields useful for correlating endpoint activity and network indicators.

02 - Process Execution Chain

Purpose: Process Execution Chain.

Splunk SPL

```
index="soc-investigation" source="*05_Comprehensive_Threat_Investigation_Raw.csv*" EventCode=4688
| table _time User ProcessName ParentProcessName ProcessId ParentProcessId CommandLine
| sort 0 _time
```

Explanation: Filters Windows Security process-creation events. ParentProcessName, ProcessId, ParentProcessId and CommandLine help reconstruct process ancestry and execution context.

03 - PowerShell Command Analysis

Purpose: PowerShell Command Analysis.

Splunk SPL

```
index="soc-investigation" source="*05_Comprehensive_Threat_Investigation_Raw.csv*" EventCode=4688
| search ProcessName="*powershell*"
| table _time User ProcessName ParentProcessName ProcessId ParentProcessId CommandLine
```

Explanation: Isolates PowerShell process-creation records and shows parent process, process identifiers and command-line arguments for static analysis.

04 - Outbound Network Connections

Purpose: Outbound Network Connections.

Splunk SPL

```
index="soc-investigation" source="*05_Comprehensive_Threat_Investigation_Raw.csv*" EventCode=3
| table _time User ProcessName DestinationIp DestinationPort Details
| sort 0 _time
```

Explanation: Displays EventCode 3 network-connection records from the simulated telemetry. The selected fields show the process and destination address/port.

05 - LSASS Process Access

Purpose: LSASS Process Access.

Splunk SPL

```
index="soc-investigation" source="*05_Comprehensive_Threat_Investigation_Raw.csv*" EventCode=10
| table _time User ProcessName TargetProcessName GrantedAccess Details
| sort 0 _time
```

Explanation: Examines Sysmon Process Access events. TargetProcessName identifies the target process, while GrantedAccess records the access mask in the event.

06 - Suspected Dump File Creation

Purpose: Suspected Dump File Creation.

Splunk SPL

```
index="soc-investigation" source="*05_Comprehensive_Threat_Investigation_Raw.csv*" EventCode=11
| search TargetFilename="*lsass*" OR TargetFilename="*debug*" OR Details="*dump*"
| table _time User ProcessName TargetFilename Details
| sort 0 _time
```

Explanation: Searches Sysmon FileCreate events for filenames or details associated with a suspected dump.

Analyst interpretation: If no results appear, inspect all EventCode 11 records and verify the field names. EventCode 11 indicates file creation, not deletion.

07 - Registry Hive Export Commands

Purpose: Registry Hive Export Commands.

Splunk SPL

```
index="soc-investigation" source="*05_Comprehensive_Threat_Investigation_Raw.csv*" EventCode=4688
| search ProcessName="*reg.exe*"
| table _time User ProcessName ParentProcessName CommandLine TargetFilename Details
| sort 0 _time
```

Explanation: Finds reg.exe process-creation events and exposes command-line arguments that may reference registry hive export/save operations.

08 - Archive and Dump Staging

Purpose: Archive and Dump Staging.

Splunk SPL

```
index="soc-investigation" source="*05_Comprehensive_Threat_Investigation_Raw.csv*"
("diagnostic_bundle.cab" OR "lsass_backup.dmp")
| table _time EventCode User ProcessName CommandLine TargetFilename Details
| sort 0 _time
```

Explanation: Correlates references to the suspected dump and archive across process and file events, ordered by time.

09 - Potential Exfiltration Correlation

Purpose: Potential Exfiltration Correlation.

Splunk SPL

```
index="soc-investigation" source="*05_Comprehensive_Threat_Investigation_Raw.csv*"
("diagnostic_bundle.cab" OR "198.51.100.99" OR "curl.exe")
| table _time EventCode User ProcessName ParentProcessName CommandLine DestinationIp DestinationPort Details
| sort 0 _time
```

Explanation: Correlates archive references, the destination address, curl.exe command lines and network events.

10 - Registry Persistence-Like Activity

Purpose: Registry Persistence-Like Activity.

Splunk SPL

```
index="soc-investigation" source="*05_Comprehensive_Threat_Investigation_Raw.csv*"
(EventCode=13 OR (EventCode=4688 AND ProcessName="*reg.exe"))
| table _time EventCode User ProcessName CommandLine RegistryKey Details
| sort 0 _time
```

Explanation: Reviews Sysmon registry-value events and reg.exe process-creation commands for possible Run-key activity.

11 - Cleanup and Log-Clearing Commands

Purpose: Cleanup and Log-Clearing Commands.

Splunk SPL

```
index="soc-investigation" source="*05_Comprehensive_Threat_Investigation_Raw.csv*" EventCode=4688
| search ProcessName="*cmd.exe" OR ProcessName="*wevtutil.exe"
| table _time User ProcessName ParentProcessName CommandLine Details
| sort 0 _time
```

Explanation: Finds command-shell and wevtutil process-creation records that may indicate cleanup or log-clearing attempts.